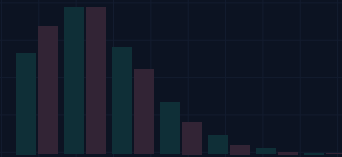


STEAMWATCH

Match Model



Dixon-Coles Probability Baselines for Football Matches

$P(H) = 45.2\%$ | $P(D) = 24.3\%$ | $P(A) = 30.5\%$

HOME
45.2%

DRAW
24.3%

AWAY
30.5%

User Guide

www.steamwatch.io



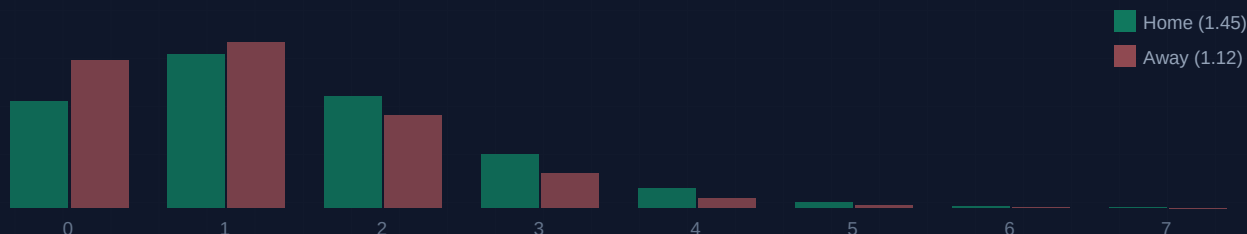
```
fair_odds = 1 / P(outcome)
l_home = atk_h * def_a * mu * hfa
tau(x,y,l1,l2,rho) = Dixon-Coles(1997)
P(X=k) = e^(-l) * l^k / k!
```

1. What Is the Match Model?

The SteamWatch Match Model uses a **Dixon-Coles adjusted Poisson regression** to estimate the probability of match outcomes (Home Win, Draw, Away Win) for football matches. It generates **fair odds baselines** that help you identify value in the betting market.

All calculations run in your browser — no data leaves your device.

Example: Poisson goal distributions for Home ($\lambda=1.45$) vs Away ($\lambda=1.12$)



2. What You Need (Data Inputs)

You enter stats for both teams. The form is split into sections — only Core Stats are required; the rest improve accuracy.

Core Stats (per team)

- Goals For / Against per match
- xG For / Against per match
- Matches Played (season)
- Penalties Received & Conceded (season totals)
- Avg Red Cards For / Against per match
- Shots For / Against per match

xG Quality Breakdown (optional but recommended)

- Open Play xG
- Set Piece xG
- Last 6 xG For / Against per match
- Last 6 xG Per Shot

Context & Adjustments

- Motivation — Normal / Elevated / Must-Win
- Absence Severity — Attack & Defence sliders (1–5 scale)
- Home Advantage Factor — pre-set per league, adjustable per match

3. Where to Find the Data

- **Opta Analyst** (theanalyst.com) — Goals, xG, open play / set piece xG breakdown
- **FBref** (fbref.com) — Shots per game, shot quality metrics

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- **Transfermarkt** ([transfermarkt.com](https://www.transfermarkt.com)) — Penalties awarded / conceded, injury & absence info
 - **Scoreroom** ([scoreroom.com](https://www.scoreroom.com)) — Red cards, discipline stats

4. How the Model Works

The model runs an 11-step pipeline. Each step refines the expected goals estimate before converting it into match probabilities.

1. Penalty xG Adjustment

Strips out penalty xG (0.76 per penalty) and reintroduces 50% to reflect actual conversion rates, giving a truer picture of open-play attacking quality.

2. Set Piece xG Discount

Applies a discount (default 85%) to set piece xG, since set pieces are less repeatable than open play chances.

3. Red Card Normalization

Adjusts attacking and defensive strength based on each team's red card frequency. More reds = weaker attack, worse defence.

4. xG Per Shot Quality

Compares each team's shot quality (xG per shot) against the league average. Teams that create higher-quality chances get a small boost.

5. Form Weighting

Blends full-season stats with the last 6 matches (default 75/25 split). Captures recent form without overreacting to small samples.

6. Attack & Defence Strength

Calculates each team's attacking and defensive ratings relative to the league average. Attack uses xG for; defence uses an 80/20 blend of xG against and actual goals against.

7. Expected Goals (Lambda)

Combines attack strength, defence strength, league average, and home advantage into a single expected goals figure per team — the Poisson λ parameter.

8. Motivation & Absences

Applies multipliers for team motivation (e.g., relegation battle = +10%) and reduces/increases expected goals based on key player absences.

9. Poisson Distribution

Uses each team's λ to generate a probability distribution for 0–10 goals, then builds an 11×11 scoreline probability matrix.

10. Dixon-Coles Correction

Adjusts the joint probability matrix to correct for the known tendency of basic Poisson models to underestimate low-scoring draws (0-0, 1-1). Uses the rho parameter (default -0.03).

11. Draw Inflation

Applies a final 8% uplift to draw probability to match real-world frequencies, then normalises all three outcomes to sum to 100%.

5. Understanding the Output

Example: Scoreline probability matrix (Dixon-Coles adjusted)

		Away Goals					
		0	1	2	3	4	5
Home Goals	0	8.0	8.2	4.8	1.8		
	1	10.7	12.8	7.0	2.6		
	2	8.0	9.0	5.0	1.9		
	3	3.9	4.4	2.4			
	4	1.4	1.6				
	5						

- **Probability %** — The model's estimated chance of Home Win, Draw, and Away Win.
- **Fair Odds** — Calculated as $1 \div \text{probability}$. If a bookmaker offers higher odds than the fair price, there may be value.
- **Expected Goals (λ)** — The Poisson rate parameter for each team. Higher λ = more goals expected.
- **Calculation Log** — Expandable breakdown showing every step, so you can see exactly how the numbers were derived.

Example output:

HOME	DRAW	AWAY
52.3%	22.1%	25.6%
Fair Odds: H 1.91 D 4.52 A 3.91		

6. Advanced Settings

For experienced users — these parameters let you fine-tune the model. The defaults work well for most matches.

Parameter	Default	What It Does
Draw Inflation	1.08	Boosts draw probability to match real-world frequencies
Dixon-Coles rho	-0.03	Controls low-score correction strength (0-0 and 1-1 draws)
Form Weight	0.25	Weight given to last 6 matches vs full season (0 = season only)
Set Piece xG Discount	0.85	Discount applied to set piece xG (1.0 = no discount)
xG/Shot Quality Weight	0.15	How much shot quality influences the estimate
Absence Weight	0.03	Impact per absence severity level on expected goals

7. Supported Leagues

League	Home Adv	Avg Goals	Avg xG	Avg Shots
Premier League	1.12	1.43	1.35	13.2
Bundesliga	1.15	1.50	1.45	14.1
La Liga	1.14	1.30	1.25	12.4
Serie A	1.13	1.32	1.26	13.0
Ligue 1	1.11	1.33	1.27	12.8
Champions League	1.18	1.45	1.40	13.5
Europa League	1.18	1.40	1.35	13.0

8. Tips

- *Always use the most recent season stats.*
- *Update last-6 form data before each matchweek.*
- *Absence severity 3+ has meaningful impact — use it for confirmed injuries to key players.*
- *Compare model fair odds to Pinnacle closing lines for the best value assessment.*
- *The model works best as a baseline — combine with your own tactical knowledge.*